

Machine Learning Poster Symposium

2025-2026 Fall

MyoPong: Goal-Conditioned Synergistic Action and Latent Exploration for Musculoskeletal Table Tennis Control with Training-Free High-Level Policies

Background

Motivation

- Bio-inspired motor systems, such as the **musculoskeletal system** (MSK), remain underexplored due to their high dimensionality and overactuated nature.



Project Goal

- We aim to tackle the **Table-Tennis task of MyoChallenge**. MSK control problem: myoArm (63 muscles) and myoTorso (210 muscles) and plus pelvis translation actuators, to control a handheld paddle and consistently return ping-pong balls launched randomly

Challenges

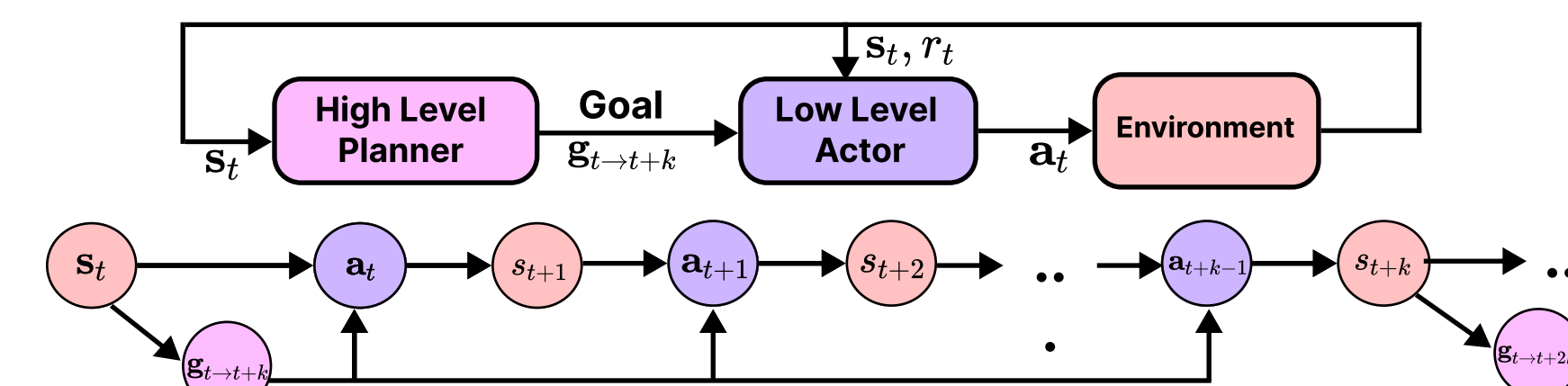
- High Dimensionality** and **Overactuated System** need efficient exploration strategy
- Table tennis environment has **sparse reward** in nature

Muscle Synergy + Latent Exploration + Goal-Oriented Hierarchical Model

Methodology

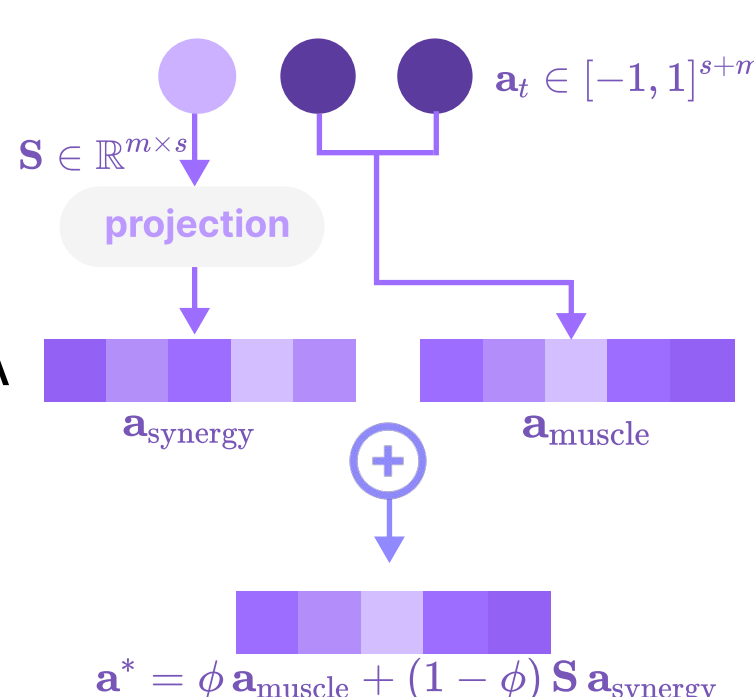
Hierarchical Model with Physics-Based Goal

- High-Level Policy guides the agent using ball kinematics analytically derived paddle position and orientation target calculated every k steps
- Low-level policy learn goal-conditioned control, trained with PPO



Muscle Synergy Extraction

- Inspired by **SAR** [1], we collected muscle activation trajectories and extract dominant co-activation patterns by applying **PCA** followed by **ICA** producing synergy matrix **S**
- Final action is calculated as linear combination of synergistic action and raw muscle action



Latent Exploration

- Using generalized time-correlated noise, inspired by the Lattice method [2]
- We add learnable perturbation matrices: a muscle-space perturbation matrix, and a latent space perturbation matrix with final action calculated as

$$a = (W + P_a + WP_x)x$$

with perturbations sampled from learnable covariant matrix

$$(P_a)_{i,j} \sim \mathcal{N}(0, (\Sigma_a)_{i,j}), \quad (P_x)_{i,j} \sim \mathcal{N}(0, (\Sigma_x)_{i,j})$$

Curriculum Learning

- The table-tennis environment is divided to two phases: (1) Phase 1 with random initial ball position and **fixed ball velocity** (2) Phase 2 with random initial ball position and **random ball velocity**
- We leveraged curriculum learning to tackle phase 2 by training the model on **increasing difficulty** from phase 1 to phase 2.

Experiment Setup

Learning Rate 1e-4 with linear decay

Network Arch Policy: 2x512; Value:2x256

Initial Log std -0.5

Clip Value 0.2

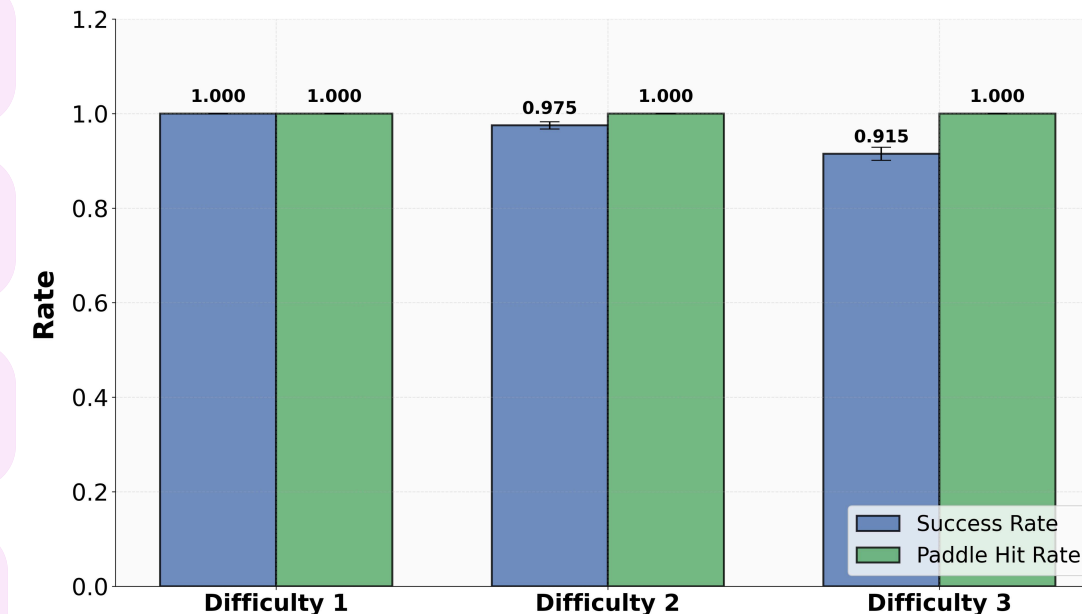
- We trained **six models**—PPO, PPO with Synergy, PPO with Lattice (Latent Exploration), and their hierarchical counterparts—using Phase 1 of the environment, which features random initial ball positions but fixed velocities.
- All baseline methods for the ablation study, as well as our proposed hierarchical methods, were trained using **PPO** based on SB3 library. Observations and rewards were normalized and clipped to stabilize learning.
- For Phase 2, we continued training only the **Hierarchical with Synergy** and **Hierarchical with Lattice** models to evaluate their adaptability to a more challenging task with random ball velocities.

Conclusion

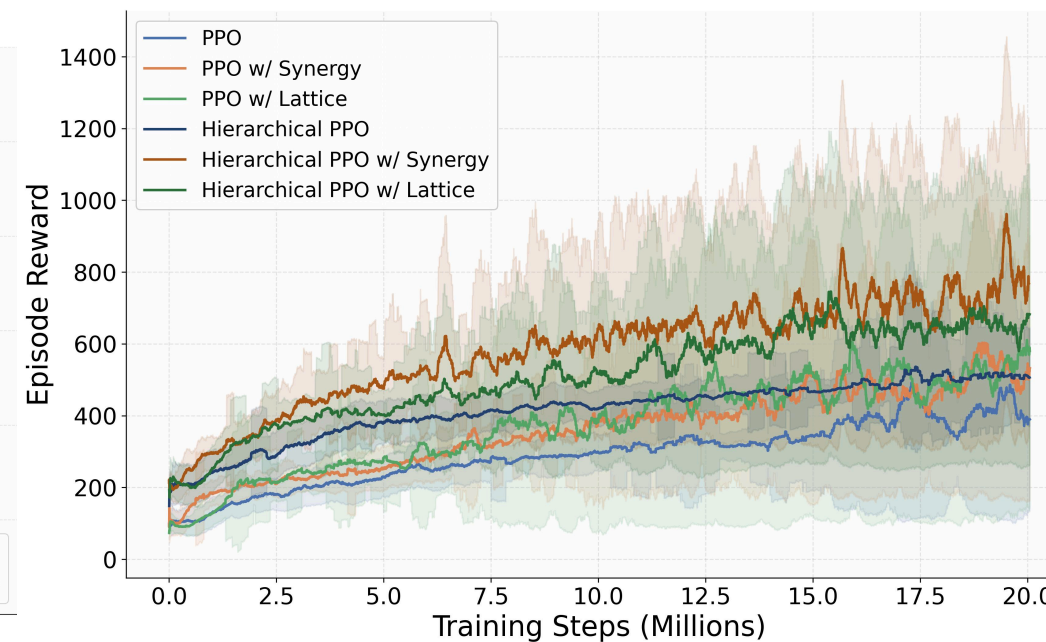
- Integrating **inductive bias on the action and exploration space** (Muscle Synergy & Lattice exploration) is critical for mastering complex, over-actuated musculoskeletal systems.
- While baseline PPO failed completely, our Hierarchical + structured exploration achieved a ~94% success rate. However, **Hierarchical architecture alone is insufficient**; it requires optimized action spaces (Synergy) or exploration (Lattice) to function effectively.
- The Lattice method proved superior to Synergy extraction, offering the best **balance of task success and energy efficiency** (minimal control effort).

Result and Findings

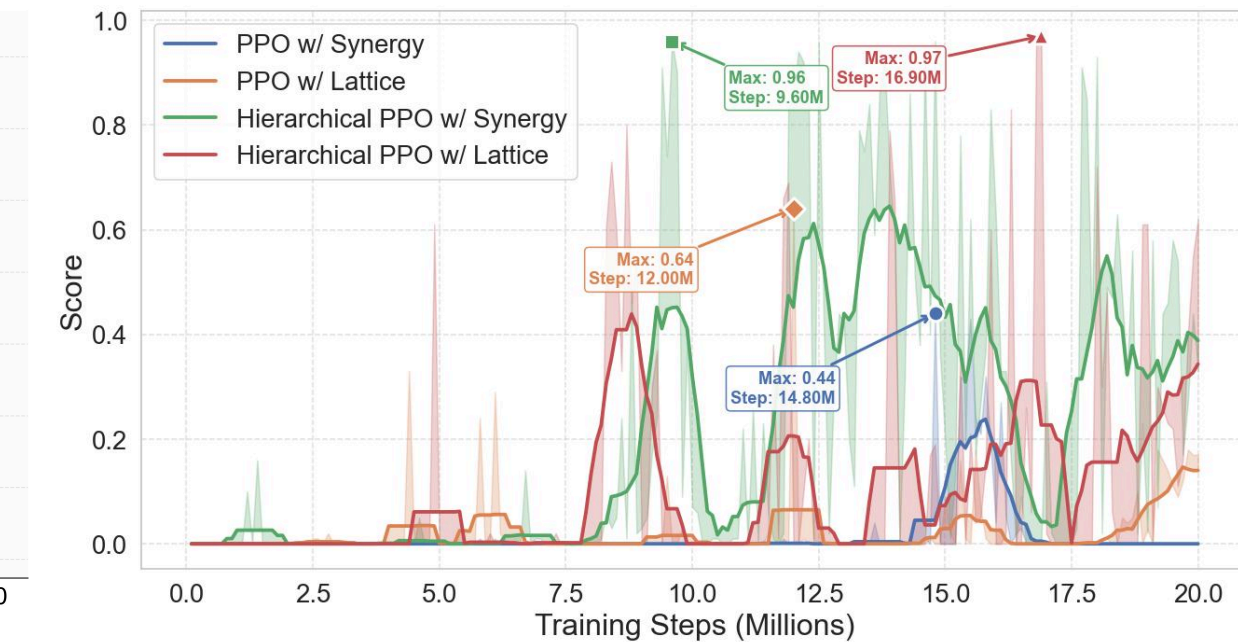
Physics-Based HLP Performance across Difficulties



Phase 1 Training

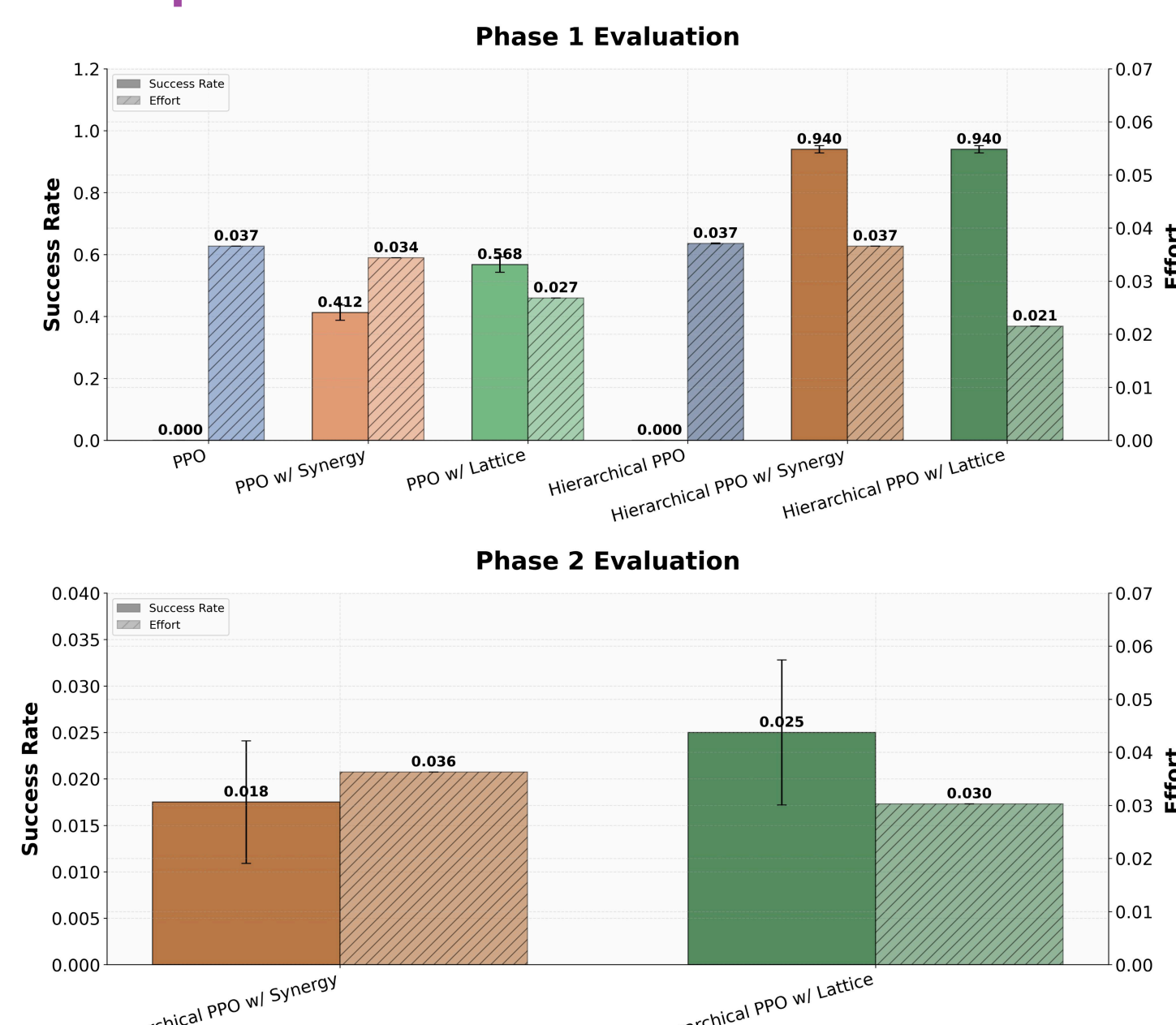


Phase 1 Training Evaluation Score Comparison



Physics-Based High-Level Policy Achieves High Accuracy on Simplified 6-DOF Paddle Control

- We tested the high-level policy in three difficulty levels, each introducing increasing stochasticity: fixed ball velocity, random ball velocity, and random friction in a simplified environment where only the paddle is controlled in 6 degrees of freedom.
- Our physics-based heuristics achieved 91.5% success on the hardest difficulty, demonstrating the **reliability and effectiveness of the high-level planner**.



Synergy Action and Latent Exploration improves baseline PPO

- Augmenting the action space with **Muscle Synergy** stabilizes the agent by reducing the action space, enabling the agent to **achieves success rate of 40%** compared to baseline PPO of 0%.
- Implementing Latent exploration improves the model more significantly achieving **56%** success rate.

Hierarchical Architecture with Synergy or Latent Exploration Achieves Near-Optimal Results

- The proposed **Hierarchical PPO architecture consistently outperforms standard PPO baselines** on both synergy and latent exploration model, achieving ~94% success rate compared to only 40%–60% for non-hierarchical variants.
- In contrast, hierarchical architectures without any optimization of the action or exploration space fail to achieve successful results.

Latent Exploration Reduces Control Effort

- While both hierarchical methods achieve high success, the **Lattice approach is the most efficient solution**; it matches the top-tier performance of Synergy but requires significantly less control effort.

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Demo available at <https://myopong.github.io>